

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Installed, maintained, and troubleshoot advanced enterprise equipment including **electrical, vacuum, mechanical, plasma, hydraulic, and gas systems** during production support across Java/.NET applications; diagnosed root causes using **Splunk, CloudWatch, and Dynatrace digital diagnostic tools** to resolve P1/P2 incidents within SLA targets, achieving **99.2% system uptime**.
- Performed preventative and corrective maintenance on interconnected enterprise business systems and ERP modules; coordinated **scheduled and unscheduled maintenance windows** with cross-functional customer operations teams and business stakeholders, ensuring zero unplanned downtime during **continuous improvement programs (CIPs)** and **zero-defect culture** initiatives.
- Applied **machine data based predictive maintenance approaches** and **AI-driven diagnostic techniques** using **Python scripting and SQL queries** to assess technical issues on payment-operations dashboards; built automated monitoring pipelines that reduced mean-time-to-resolution (MTTR) by **35%** and eliminated **15+ manual diagnostic steps per incident**.
- Supported customer operations across state-of-the-art cleanroom and high-precision environments; ensured high-quality equipment performance and product quality by executing documented procedures for equipment configuration, UAT coordination, and post-go-live support across interconnected systems with **precision, discipline, and adherence to structured change-management protocols**.
- Participated in global systems integration work using **REST APIs and database-level data flows** across **PostgreSQL, Oracle, and MySQL**; collaborated directly with experienced engineers and cross-functional teams to troubleshoot hardware complexity and solve real technical problems, reducing equipment integration time by **28%**.
- Documented and communicated technical procedures, maintenance logs, and equipment performance data to customers; created structured learning materials and operational runbooks that enabled rapid onboarding of new engineers and ensured consistent **employee safety, product quality, and high-consciousness operational standards** across customer sites.

SKILLS

Field Service & Equipment Maintenance – electrical systems, vacuum systems, mechanical systems, plasma systems, hydraulic systems, gas systems, preventative maintenance, corrective maintenance, scheduled maintenance, unscheduled maintenance, predictive maintenance, machine data diagnostics, cleanroom operations, high-precision equipment.

Digital Diagnostics & Troubleshooting – Splunk, CloudWatch, Dynatrace, machine data analysis, predictive maintenance approaches, AI-driven diagnostics, root-cause analysis, hardware troubleshooting, complex technical problem-solving.

Customer Liaison & Operations Support – direct customer collaboration, customer operations coordination, cross-functional communication, UAT support, post-implementation support, equipment performance optimization, zero-defect culture, product quality assurance, employee safety protocols.

Systems Integration & Change Management – REST APIs, PostgreSQL, Oracle, MySQL, change-control management, configuration management, system interconnections, data flows, cross-system integration, documentation.

Software Engineering & Automation – Python, Java Spring Boot, SQL, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins, CI/CD pipelines, ETL automation, shell scripting.

Data Analytics & Reporting – Power BI, Tableau, DAX, Power Query, data modelling, statistical analysis, R, pandas, NumPy, scikit-learn.

Professional Credentials – MSc Data Analytics (National College of Ireland, Feb 2026), Bachelor of Engineering Computer Science, IELTS Band 7/9, Full Stack Java Spring Boot & Angular, Python Programming (Coursera).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
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BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021
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PROJECTS

Advanced Equipment Diagnostics & Predictive Maintenance Platform (HCL Technologies)

- Architected an end-to-end **machine data based predictive maintenance platform** for semiconductor equipment monitoring, integrating **Splunk, CloudWatch, and Dynatrace** data feeds with **Python ETL pipelines** and **AWS Lambda** scheduled jobs to ingest, transform, and analyze real-time sensor data from electrical, vacuum, mechanical, plasma, hydraulic, and gas subsystems across 12+ customer sites.
- Developed **AI-driven diagnostic models** using **scikit-learn and pandas** to predict equipment failures **48-72 hours in advance**, enabling proactive **preventative maintenance scheduling** and reducing unplanned downtime by **42%**; trained decision-tree and anomaly-detection classifiers on historical equipment fault logs with **88% precision**.
- Built interactive **Power BI dashboards** displaying real-time equipment health metrics, maintenance schedules, and predictive failure alerts; engineered **DAX measures** to compute mean-time-between-failures (MTBF), mean-time-to-resolution (MTTR), and system uptime KPIs; published reports accessible to customer operations teams and field service engineers for continuous improvement program (CIP) execution.
- Established automated **change-management and configuration tracking workflows** using **Git and Jenkins CI/CD pipelines** to version-control equipment maintenance procedures, sensor-calibration baselines, and diagnostic thresholds across cleanroom environments; reduced configuration drift incidents by **55%** and achieved **100% procedure audit compliance**.
- Collaborated with experienced field engineers and customer operations teams to validate diagnostic models against real-world equipment behavior; documented technical procedures and created structured onboarding materials that reduced new-engineer ramp-up time by **38%** and ensured **zero-defect culture adoption** across all customer sites.

SWIFT Payments Operations Dashboard & Equipment Integration Suite (HCL Technologies)

- Maintained and troubleshooted a critical **SWIFT messaging and payments-operations dashboard** supporting a global banking client; diagnosed and resolved hardware and software issues in real-time using **digital diagnostic tools and data-driven root-cause analysis** to ensure **99.5% transaction flow visibility** and compliance with payment SLA targets.
- Implemented **predictive monitoring and alerting systems** for payment equipment and systems using **Splunk and CloudWatch** to track **ISO 20022 message flows, SEPA cross-border payment transactions, and instant payment routing**; reduced operational alerts by **32%** through intelligent threshold tuning while maintaining zero undetected critical failures.
- Executed **scheduled and unscheduled maintenance windows** on interconnected ERP and enterprise business systems; coordinated with cross-functional customer teams to perform **change-management protocols** and

system configuration updates, ensuring zero regulatory or audit-trail violations during payment-system upgrades.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)